



Adrenal hyperandrogenism

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Literature review current through: **Jun 2026**.
 This topic last updated: **Jun 29, 2026**.

INTRODUCTION

The [adrenal glands](#) are a prominent source of [androgen](#), particularly in children of both sexes and female adults. Excess [adrenal androgen](#) secretion occurs in several different patient groups:

- It is a common and well-recognized cause of [virilization](#) in [infants](#) and children and an occasional cause of menstrual irregularities, [hirsutism](#), and [virilization](#) in female adults. (See "[Adrenal steroid biosynthesis](#)" and "[Pathophysiology and causes of hirsutism](#)".)
- It can occur in male adults, in whom it has few clinical manifestations.
- It may contribute to the clinical findings in some patients with [Cushing syndrome](#). (See "[Epidemiology and clinical manifestations of Cushing syndrome](#)".)

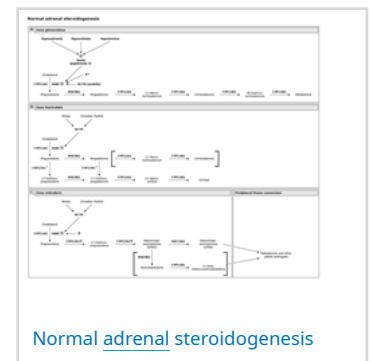
This topic will provide an overview of [adrenal androgen synthesis](#) and disorders associated with excess [adrenal androgen](#) secretion. More detailed discussions of individual disorders are found separately.

- (See "[Premature adrenarche](#)".)
- (See "[Precocious puberty: Diagnosis and evaluation](#)".)
- (See "[Classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in infants and children: Clinical manifestations and diagnosis](#)".)
- (See "[Clinical manifestations of classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in adults](#)".)
- (See "[Nonclassic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in children and adolescents](#)".)
- (See "[Nonclassic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in adults](#)".)
- (See "[Uncommon congenital adrenal hyperplasias](#)".)
- (See "[Clinical manifestations and diagnosis of adrenocortical carcinoma](#)".)

NORMAL ADRENAL ANDROGEN SYNTHESIS

Androgens are "byproducts" of [cortisol synthesis](#) in the [adrenal glands](#) [1]. As a result, the major determinant of their production is [corticotropin \(ACTH\)](#). The primary [adrenal androgens](#) are [dehydroepiandrosterone \(DHEA\)](#) and [DHEA sulfate \(DHEAS\)](#) ([figure 1](#)). Less than 10 percent of [DHEA](#) and [DHEAS](#) are produced by the [testes](#) or [ovaries](#).

[DHEA](#) and [DHEAS](#) have little, if any, intrinsic androgenic activity. However, small amounts are converted to [androstenedione](#) and then to [testosterone](#) (and to [estradiol](#)) in both the [adrenal glands](#) and [peripheral tissues](#), including [hair follicles](#), [sebaceous glands](#), the [prostate](#), external genitalia, and [adipose tissue](#). Thus, while excess secretion of [DHEA](#) and [DHEAS](#) defines [adrenal hyperandrogenism](#), the menstrual irregularities, [hirsutism](#), and [virilization](#) are in fact caused by the more potent [androgens androstenedione](#) and [testosterone](#). [DHEA](#) and [DHEAS](#) are metabolized primarily by reduction and conjugation to sulfates or glucuronides in the liver; the metabolites of these [steroids](#) and [androstenedione](#) are excreted as 17-ketosteroids in the urine.



Normal [adrenal steroidogenesis](#)

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Adrenal hyperandrogenism - UpToDate

Figure 1 - larger image below

DHEA and DHEAS are secreted in small amounts during [infancy](#) and early childhood. Their secretion gradually increases during late childhood, and the increase correlates with the development of small amounts of pubic and/or axillary hair in older, but still prepubertal, children, a phenomenon known as [adrenarche](#). The cause of this increase in [androgen](#) secretion is not known; however, the increase is in parallel with the development of the [zona reticularis](#) of the [adrenal cortex](#). ACTH and [cortisol](#) secretion do not increase, suggesting that ACTH is not responsible. Other [proopiomelanocortin \(POMC\)](#)-derived [peptides](#) or a postulated, but never confirmed, [adrenal androgen-stimulating hormone](#) may be involved [2,3]. Although the major factor responsible for [adrenarche](#) is unknown, several hypotheses are discussed separately. (See "[Physiology and clinical manifestations of normal adrenarche](#)", [section on 'Regulation of zona reticularis/adrenarchal growth, development, and function'](#).)

In male adults, less than 5 percent of [testosterone](#) is derived from the [adrenal glands](#) or from [adrenal](#)-derived precursors in peripheral tissues. In female adults, however, [androstenedione](#) and [testosterone](#) secreted by the [adrenals](#) and produced peripherally from [DHEA](#), and to a lesser extent from DHEAS, contribute substantially to total [testosterone](#) production. In the [follicular phase of the menstrual cycle](#), for example, the [adrenals](#) directly or indirectly account for approximately 65 percent of [testosterone](#) production. In the midcycle, the ovarian contribution increases, so that the [adrenals](#) contribute only approximately 40 percent.

In both males and females, serum [DHEA](#) and DHEAS concentrations rise throughout [puberty](#) and for a few years thereafter, reaching a peak in the third decade of life (the concentrations of DHEAS are 100 to 1000 times higher than those of [DHEA](#) at all ages). Thereafter, they decline progressively; by age 80 years, the concentrations are only approximately 25 percent of those at age 25 years [4]. This decline is sometimes called andropause, and neither its cause nor potential consequences have been established.

CLINICAL MANIFESTATIONS

The clinical manifestations of adrenal hyperandrogenism vary with the age at onset and the sex of the patient.

Prepubertal children — In prepubertal children, androgen excess from any source increases height velocity, somatic development, and skeletal maturation. Epiphyseal fusion may occur prematurely, leading to short adult height. (See "[Premature adrenarche](#)" and "[Precocious puberty: Diagnosis and evaluation](#)".)

Other findings vary with the sex of the child:

- In prepubertal boys, increases in androgen exposure causes virilization manifested by penile enlargement, growth of hair in androgen-dependent areas, deepening of the voice, and development of other secondary sexual characteristics (eg, isosexual precocious puberty)
- In prepubertal girls, androgen excess causes hirsutism, acne, and clitoromegaly (eg, contrasexual precocious puberty)

Pubertal children — In pubertal boys, androgen excess of any cause increases the rate of progression of puberty and skeletal maturation, which can lead to premature epiphyseal fusion, thereby decreasing adult height. However, pubertal development and growth may also be arrested in boys who have concurrent hypercortisolism (Cushing syndrome) [5] because chronic cortisol excess inhibits pituitary-gonadal function and growth hormone secretion [6]. (See "[Children and adolescents with short stature: Diagnostic approach](#)" and "[Precocious puberty: Diagnosis and evaluation](#)".)

In pubertal girls, androgen excess causes virilization, primary or secondary amenorrhea, and increased skeletal maturation. As in boys, concurrent hypercortisolism may cause gonadal suppression and stunt linear growth.

Adults — In male adults, adrenal androgen excess has little effect (ie, hair growth or muscle mass do not increase; however, acne and hirsutism in early puberty may occur). It does, however, inhibit gonadotropin secretion so that testes size, testicular testosterone secretion, and spermatogenesis may decrease [7].

In female adults, increased adrenal androgen production causes hirsutism, acne, male-pattern baldness, menstrual irregularities, oligomenorrhea or amenorrhea, infertility, and even frank virilization. (See "[Clinical manifestations of classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in adults](#)", [section on 'Female reproduction'](#) and "[Evaluation and management of postmenopausal hyperandrogenism](#)".)



PRIMARY ADRENAL CAUSES

Excessive adrenal production of androgens may occur as a result of both acquired and inherited adrenal diseases and other diseases that affect adrenal function ([table 1](#)).

Premature adrenarche — Premature adrenarche is the appearance of pubic or axillary hair before the age of eight years in girls and nine years in boys, without other signs of puberty or virilization and

Causes of adrenal hyperandrogenism	
Primary adrenal diseases	
Premature adrenarche	
Adrenal tumors (adenomas, carcinomas, bilateral macronodular adrenal hyperplasia)	
ACTH hypersecretion	
Congenital adrenal hyperplasia	
ACTH-dependent Cushing's syndrome	
Glucocorticoid resistance	
Cortisone reductase deficiency	

without a major advance in bone age [8]. It is more common in girls than in boys, usually occurs after the age of six years, and is not progressive. (See "Premature adrenarche".)

A rare genetic defect in dehydroepiandrosterone (DHEA) sulfation, caused by a pathogenic variant in the gene encoding the sulfate donor PAPSS2, has been described. This variant results in premature pubarche (isolated premature appearance of sexual hair), hirsutism, acne, oligomenorrhea, and very low (undetectable) DHEA sulfate (DHEAS) levels. (See "Uncommon congenital adrenal hyperplasias", section on 'PAPSS2 deficiency (apparent DHEA sulfotransferase deficiency)').

Hyperandrogenism	
Exogenous	
Androgens	
Pregnancy	
Placental sulfatase deficiency (no hyperandrogenism)	
Placental aromatase deficiency (mother and female fetus virilized)	
P450 oxidoreductase deficiency	
Other	
PAPSS2 deficiency (apparent DHEA sulfotransferase deficiency)	

Causes of adrenal hyperandrogenism

Table 1 - larger image below

Adrenal tumors — Both benign and malignant adrenal tumors or bilateral macronodular adrenal hyperplasia can produce mainly or only androgens, including testosterone.

Adenomas — Androgen-secreting adrenal **adenomas** are rare. Affected patients have high serum androgen concentrations that do not fall in response to dexamethasone; in rare cases, the tumor secretes only testosterone [9]. These patients usually do not have marked clinical or biochemical androgen excess. The tumors are typically small (diameter less than 4 cm) but are generally visible on computed tomography (CT) or magnetic resonance imaging (MRI).

Carcinomas — Androgen-secreting adrenal **carcinomas** are more common than adenomas. These tumors characteristically produce androgens, many steroid intermediates, and sometimes cortisol. Some patients have a palpable abdominal mass or metastatic disease at the time of diagnosis, and many have clinical manifestations of both cortisol and androgen excess. Nearly all patients have very high serum DHEA and DHEAS concentrations and urinary 17-ketosteroid excretion, and the values do not fall in response to high-dose dexamethasone. Serum cortisol concentrations and urinary cortisol excretion are often increased. Most adrenal carcinomas are larger than 5 cm in diameter and have already invaded the capsule of the gland or neighboring tissues by the time they are discovered. (See "Clinical manifestations and diagnosis of adrenocortical carcinoma", section on 'Clinical presentation'.)

Bilateral macronodular adrenal hyperplasia — Although the majority of patients with bilateral macronodular adrenal hyperplasia present with subclinical or overt Cushing syndrome, some have cosecretion of cortisol with other steroids, including adrenal androgens. (See "Cushing syndrome due to primary bilateral macronodular adrenal hyperplasia".)

ACTH-DEPENDENT CAUSES

There are several forms of adrenal hyperandrogenism that are mediated by corticotropin (ACTH).

Congenital adrenal hyperplasia — Congenital adrenal hyperplasia is the prototypic disorder of adrenal androgen secretion. Hyperandrogenism occurs in those disorders in which cortisol synthesis is impaired, but the androgen secretory pathway is intact. These include CYP21A2 (21-hydroxylase) and CYP11B1 (11-beta-hydroxylase) deficiencies. (See "Clinical manifestations of classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in adults" and "Uncommon congenital adrenal hyperplasias".)

Hexose-6-phosphate-dehydrogenase deficiency (also called apparent cortisone reductase deficiency) leads to diminished activity of 11-beta-hydroxysteroid dehydrogenase type 1 [10]. The type 1 enzyme converts the inactive cortisone to cortisol in the liver and other tissues. Deficiency of this enzyme is compensated for by ACTH hypersecretion in a mechanism similar to that described for glucocorticoid resistance. This disorder is discussed in greater detail separately. (See "Uncommon congenital adrenal hyperplasias", section on 'Hexose-6-phosphate-dehydrogenase deficiency (apparent cortisone reductase deficiency)').

In P450 oxidoreductase deficiency (ORD), affected neonates may present with severe undervirilization in boys and severe virilization in girls [10]. Maternal virilization, associated with a high excretion of androgen metabolites beginning around midgestation, is frequently but not invariably observed during pregnancies with ORD. This disorder is reviewed in detail separately. (See "Uncommon congenital adrenal hyperplasias", section on 'P450 oxidoreductase deficiency (apparent combined CYP17A1 and CYP21A2 deficiency)').

ACTH-dependent Cushing syndrome — Corticotropin (ACTH)-dependent Cushing syndrome is the most common cause of endogenous hypercortisolism, most often caused by a corticotroph adenoma of the pituitary, occasionally by ectopic ACTH secretion, and very rarely by excess corticotropin-releasing hormone (CRH) secretion. Thus, cortisol secretion should be evaluated in any adult patient with androgen excess. (See "Establishing the diagnosis of Cushing syndrome" and "Establishing the cause of Cushing syndrome".)

Primary familial or sporadic glucocorticoid resistance syndrome — The rare syndrome of familial or sporadic glucocorticoid resistance is due to inactivating variants in the gene for the glucocorticoid receptor and results in partial resistance to the actions of cortisol in all tissues [7]. As a result of pituitary resistance, there are increases in ACTH secretion and, therefore, in the secretion of cortisol, adrenal androgens, and the mineralocorticoid 11-deoxycorticosterone. The high serum cortisol concentrations maintain normal or near-normal glucocorticoid activity in tissues. Thus, the manifestations of glucocorticoid resistance vary from none to chronic fatigue (perhaps reflecting glucocorticoid deficiency), to

variable degrees of hyperandrogenism or mineralocorticoid excess. (See ["Primary adrenal insufficiency in children: Causes"](#), section on 'End-organ unresponsiveness'.)

DIFFERENTIAL DIAGNOSIS

Hypersecretion of androgens from the adrenal glands nearly always results in high serum dehydroepiandrosterone (DHEA) and DHEA sulfate (DHEAS) concentrations, and one or both of these compounds (and serum testosterone) should be measured in any hirsute or virilized woman in whom the cause is not obvious [11,12]. The results of serum DHEA and DHEAS measurements must be interpreted based on the patient's age. A high DHEA or DHEAS value in a hirsute or virilized patient strongly suggests adrenal hyperandrogenism, whereas serum testosterone values are high in patients with either adrenal or ovarian hyperandrogenism [13,14]. (See ["Evaluation of premenopausal females with hirsutism"](#) and ["Evaluation and management of postmenopausal hyperandrogenism"](#).)

Most androgen-secreting adrenal tumors can be identified by computed tomography (CT) or magnetic resonance imaging (MRI). Adrenal masses with low signal intensity on T1- and T2-weighted MRI images are usually adenomas, whereas carcinomas have low signal intensity on T1-weighted images and enhanced activity on T2-weighted images ([image 1](#) and [image 2](#)). (See ["Evaluation and management of the adrenal incidentaloma"](#), section on 'Evaluation for malignancy'.)

However, adrenal hyperandrogenism is not the only cause of elevated serum levels of DHEA or DHEAs. Other potential causes include the following:

Hyperprolactinemia — Rarely, women with marked hyperprolactinemia due to a lactotroph adenoma have slight hirsutism and slightly elevated serum dehydroepiandrosterone (DHEA), DHEA sulfate (DHEAS), and androstenedione concentrations [15,16]. The cause of the hyperandrogenism is uncertain. Serum prolactin need not be measured in female adults with hirsutism, unless they have other manifestations of hyperprolactinemia such as oligomenorrhea, amenorrhea, or especially galactorrhea. (See ["Clinical manifestations and evaluation of hyperprolactinemia"](#).)

The treatment of hyperprolactinemia varies with the cause. (See ["Management of hyperprolactinemia"](#).)

Exogenous DHEA intake — Dehydroepiandrosterone (DHEA) has been studied as a potential therapy for a number of disorders, including adrenal insufficiency, depression, menopausal symptoms, fibromyalgia, and as a possible antiaging agent. However, the clinical benefits and safety of DHEA for any clinical indication are not well established. While DHEA is available by prescription only in most countries, in the United States, DHEA is widely available in stores that sell health foods and nutritional supplements. However, quality control (purity and potency) of these over-the-counter products has been shown to be quite poor. (See ["Treatment of adrenal insufficiency in adults"](#), section on 'Approach to a therapeutic trial'.)

Placental aromatase deficiency — Pregnant patients who have placental aromatase or sulfatase deficiency are unable to produce estradiol and estriol from their precursors, DHEAS and 16-alpha-DHEAS, which come from the fetus [17,18]. Mothers with placental aromatase deficiency and their female fetuses may have some signs of androgen excess and increases in serum DHEA, DHEAS, and testosterone concentrations. In comparison, pregnant patients with placental sulfatase deficiency are asymptomatic but have low serum estriol concentrations. (See ["Gestational hyperandrogenism"](#).)

Pregnant patients with placental sulfatase deficiency need no therapy. Those with aromatase deficiency who have female fetuses could be treated with [dexamethasone](#), which crosses the placenta and inhibits fetal adrenal function. However, the efficacy of this approach is not known. (See ["Gestational hyperandrogenism"](#), section on 'Placental aromatase deficiency'.)

TREATMENT AND PROGNOSIS

The treatment and prognosis of patients with adrenal hyperandrogenism vary with the underlying cause.

Premature adrenarche — No treatment is needed besides reassurance. Pubertal development usually begins at the expected time. When associated with obesity or insulin resistance, appropriate advice for diet and lifestyle changes should be given [19]. (See ["Premature adrenarche"](#).)

 **en associated with nonclassic congenital adrenal hyperplasia**, appropriate glucocorticoid replacement is given. (See ["Nonclassic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in children and adolescents"](#), section on 'Children'.)

Adrenal tumors — Surgery is the treatment of choice for all patients with hormone-secreting adrenal tumors [20,21]. Surgery is the initial treatment for patients with adrenocortical carcinoma and is technically feasible in most. (See ["Treatment of adrenocortical carcinoma"](#).)

Congenital adrenal hyperplasia — Patients with congenital adrenal hyperplasia are treated with a glucocorticoid and often a mineralocorticoid. (See ["Classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in infants and children: Treatment"](#) and ["Treatment of classic](#)

congenital adrenal hyperplasia due to 21-hydroxylase deficiency in adults" and "Nonclassic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in children and adolescents" and "Uncommon congenital adrenal hyperplasias".)

ACTH-dependent Cushing syndrome — Patients with corticotropin (ACTH)-dependent Cushing syndrome due to a corticotroph adenoma should be treated with pituitary surgery. Patients with ectopic ACTH or corticotropin-releasing hormone (CRH)-secreting tumors are more difficult to treat. (See "Overview of the treatment of Cushing syndrome" and "Primary therapy of Cushing disease: Transsphenoidal surgery and pituitary irradiation".)

Primary glucocorticoid resistance — Patients with glucocorticoid resistance should be treated with appropriately titrated doses of a glucocorticoid that has no intrinsic mineralocorticoid activity, such as [dexamethasone](#) [7]. The goals of therapy are to reduce ACTH and therefore adrenal androgen (and mineralocorticoid) secretion, taking care not to cause iatrogenic Cushing syndrome. Women can be given [spironolactone](#) or [flutamide](#) to block the action of androgen. (See "Management of hirsutism in premenopausal females" and "Primary adrenal insufficiency in children: Causes", section on 'End-organ unresponsiveness'.)

SUMMARY

- **Adrenal androgens** – The adrenal glands are a prominent source of androgens. In fact, dehydroepiandrosterone (DHEA) and DHEA sulfate (DHEAS) are the most abundant products of the adrenal glands. The major determinant of their production is corticotropin (ACTH) ([figure 1](#)). Less than 10 percent of DHEA and DHEAS are produced by the testes or ovaries.

DHEA and DHEAS have little, if any, intrinsic androgenic activity. However, small amounts are converted to androstenedione and then to testosterone (and to estrogen) in both the adrenal glands and peripheral tissues, including hair follicles, sebaceous glands, the prostate, external genitalia, and adipose tissue. Thus, while excess secretion of DHEA and DHEAS defines adrenal hyperandrogenism, the hirsutism and virilization are, in fact, caused by the more potent androgens androstenedione and testosterone.

- **Primary adrenal causes of adrenal hyperandrogenism**

- Premature adrenarche. (See "Premature adrenarche".)
- Adrenal tumors – Adenomas and carcinomas, and bilateral macronodular adrenal hyperplasia ([table 1](#)). (See "Clinical manifestations and diagnosis of adrenocortical carcinoma".)

- **ACTH-dependent causes of adrenal hyperandrogenism**

- Congenital adrenal hyperplasia (common forms such as 21-hydroxylase deficiency, as well as others). (See 'Congenital adrenal hyperplasia' above.)
- ACTH-dependent Cushing syndrome. (See 'ACTH-dependent Cushing syndrome' above.)
- Primary glucocorticoid resistance syndrome. (See 'Primary familial or sporadic glucocorticoid resistance syndrome' above.)

- **Differential diagnosis for elevated DHEA and DHEAS levels**

- Hyperprolactinemia. (See 'Hyperprolactinemia' above.)
- Exogenous DHEA intake. (See 'Exogenous DHEA intake' above.)
- Placental aromatase deficiency. (See 'Placental aromatase deficiency' above.)

- **Management** – Management of adrenal hyperandrogenism depends upon the underlying cause. (See 'Treatment and prognosis' above.)

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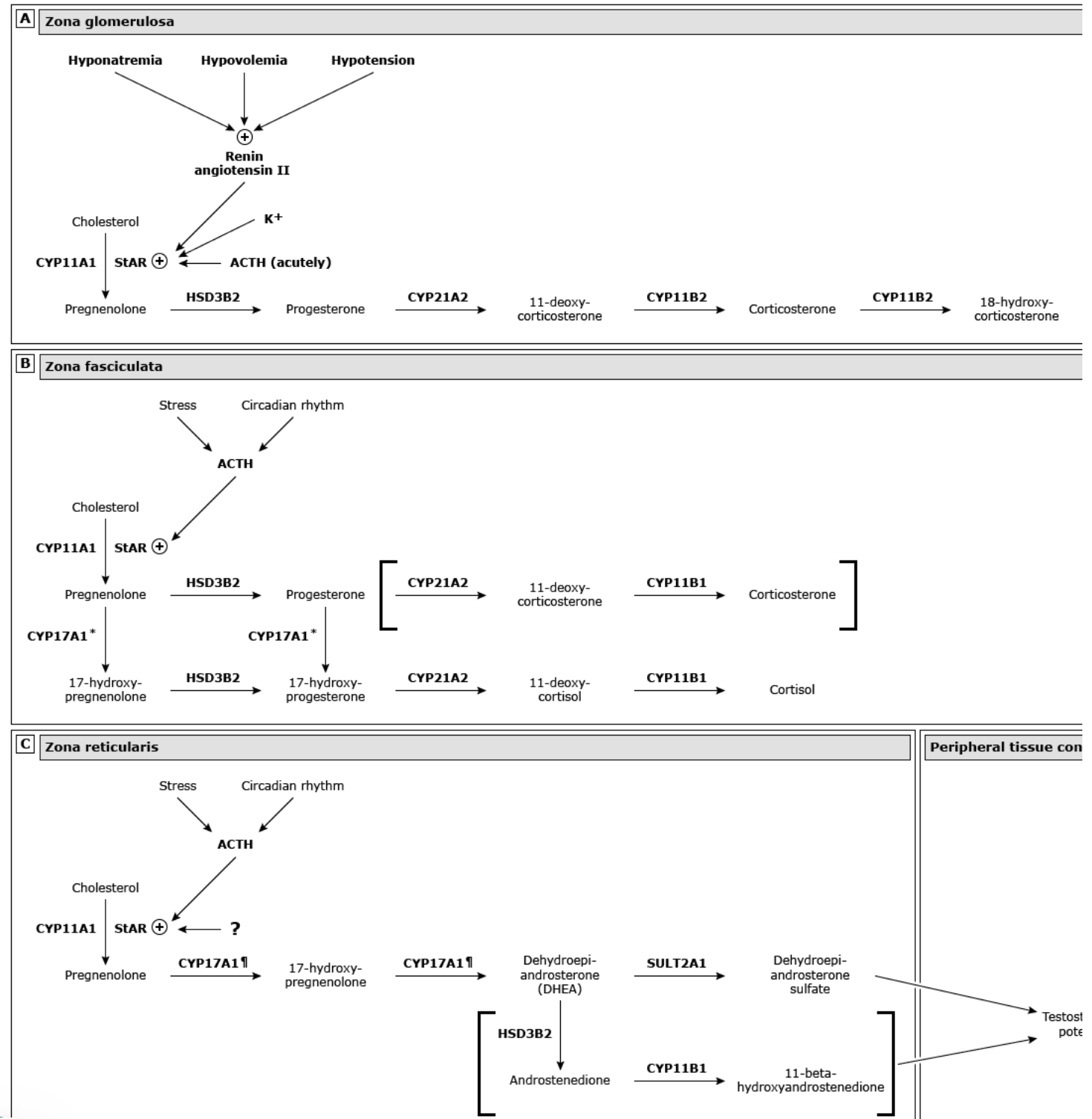
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GRAPHICS

Figure 1: Normal adrenal steroidogenesis



(A) Zona glomerulosa (ZG): Controlled primarily by the renin-angiotensin system through angiotensin II as well as potassium ion and ACTH (acutely). Renin secretion is stimulated by hyponatremia, hypovolemia, and hypotension. The ultimate increase in angiotensin II causes vasoconstriction and stimulates aldosterone, thereby increasing renal sodium reabsorption resulting in an expansion of plasma volume.

(B) Zona fasciculata (ZF): Cortisol is the major product because CYP17A1 (17-hydroxylase activity*) predominates. The square brackets ([]) indicate that corticosterone synthesized from progesterone by CYP21A2 and then CYP11B1 is usually a minor pathway (except in CYP17A1 [17-hydroxylase*] deficiency).

(C) Zona reticularis (ZR): The question mark (?) indicates that other factors that are not yet defined might be involved in the control of steroidogenesis in the ZR. The square brackets ([]) indicate that HSD3B2 production of androstenedione is a minor pathway in the ZR. The relatively weak androgens produced in the ZR can be converted to more potent androgens in peripheral tissue. Refer to other UpToDate graphic content on testosterone formation and metabolism.

ACTH: corticotropin.

* CYP17A1 has high 17-hydroxylase activity in the zona fasciculata but has minimal 17,20-lyase activity because of low expression of a cofactor (cytochrome b5) necessary for full 17,20-lyase activity. Therefore, the ZF is not a source of significant adrenal androgen precursors as in the ZR (C).

¶ In the ZR, CYP17A1 catalyzes 17-hydroxylase and 17,20-lyase activity because of the presence of a cofactor (cytochrome b5).

Courtesy of Hershel Raff, PhD.

Graphic 71558 Version 9.0

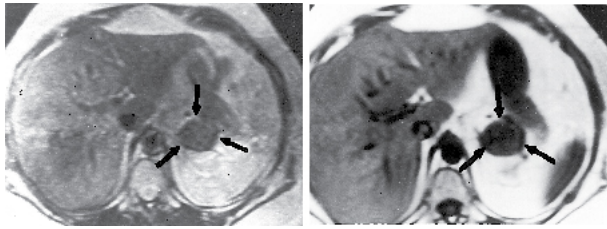
Table 1: Causes of adrenal hyperandrogenism

Primary adrenal diseases
Premature adrenarche
Adrenal tumors (adenomas, carcinomas, bilateral macronodular adrenal hyperplasia)
ACTH hypersecretion
Congenital adrenal hyperplasia
ACTH-dependent Cushing's syndrome
Glucocorticoid resistance
Cortisone reductase deficiency
Hyperprolactinemia
Exogenous
Androgens
Pregnancy
Placental sulfatase deficiency (no hyperandrogenism)
Placental aromatase deficiency (mother and female fetus virilization)
P450 oxidoreductase deficiency
Other
PAPSS2 deficiency (apparent DHEA sulfotransferase deficiency)

ACTH: corticotropin; DHEA: dehydroepiandrosterone.

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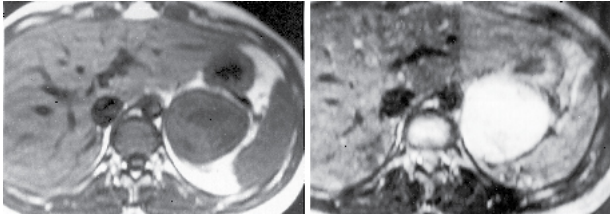


Image 1: Adrenocortical adenoma

Magnetic resonance images of an adrenocortical adenoma. Left: T1-weighted scan showing a left adrenal mass with low signal intensity similar to that of liver tissue. Right: The low signal intensity of this mass did not change during the T2-weighted scan.

Courtesy of George P Chrousos, MD.

Graphic 65020 Version 2.0

Image 2: Adrenocortical carcinoma

Magnetic resonance images of a functioning adrenocortical carcinoma. Left: T1-weighted scan showing large left adrenal mass with low signal intensity similar to that of liver tissue. Right: T2-weighted scan showing enhanced signal intensity of the same mass.

Courtesy of George P Chrousos, MD.

Graphic 62315 Version 3.0

Contributor Disclosures

George P Chrousos, MD No relevant financial relationship(s) with ineligible companies to disclose. **André Lacroix, MD** Patent Holder: Université Paris-Saclay [Endocrine diseases related to KDM1A]. Grant/Research/Clinical Trial Support: Corcept Therapeutics [Cushing's syndrome]. Consultant/Advisory Boards: Pfizer [Pituitary tumors]; Recordati [Cushing's syndrome]. Speaker's Bureau: Medunik Canada [Adrenocortical cancer]; Recordati [Cushing's syndrome]. All of the relevant financial relationships listed have been mitigated. **Xin He, MD** No relevant financial relationship(s) with ineligible companies to disclose.

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